UNIVERSIDAD DE LA SABANA

Big Data Tools – Coterminal Ingeniería Informática.

Tutorial 1: Install Miniconda

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What is Miniconda?

Miniconda is a lightweight version of Anaconda that includes only the essential Conda package manager and Python. It allows you to create isolated Python environments and manage packages efficiently without the bulk of a full Anaconda install.

Step 1: Download Miniconda Installer

- Go to the official Miniconda download page, usually downloading files with the pattern:
    - Windows (64-bit): Miniconda3-latest-Windows-x86_64.exe
    - macOS: Miniconda3-latest-MacOSX-x86_64.sh or, for Apple Silicon, Miniconda3-latest-MacOSX-arm64.sh
    - Linux (x86_64): Miniconda3-latest-Linux-x86_64.sh

Step 2: Install Miniconda

On Windows:

- Download the .exe installer.
- Run the installer and follow the prompts.
- Choose to add Miniconda to your system PATH if you want to use conda commands directly in Command Prompt or PowerShell.
- After installation, open Anaconda Prompt (or your Command Prompt if PATH added) to use Conda.

On macOS/Linux:

- Open a terminal.
- Navigate to the download folder.
- Run the installer with this command (replace filename if different):

```
bash Miniconda3-latest-Linux-x86_64.sh
```

- Follow the prompts, accept license, and set install location when asked.
- After installation, activate Conda with:

```
source ~/miniconda3/bin/activate
```

- Initialize Conda for your shell:

```
conda init
```

- Restart your terminal or source your shell config file to enable.

Step 3: Verify Installation

Run in terminal or command prompt:

```
conda --version
```

You should see the installed Conda version.

Step 4: Use Conda to Manage Environments and Packages

- To create a new Python environment (e.g., Python 3.10):

```
conda create --name bigdata python=3.11
```

- Activate the environment:

```
conda activate bigdata
```

- Install packages, for example, pandas:

```
conda install pandas
```

- List installed packages:

```
conda list
```

- Deactivate the environment:

```
conda deactivate
```